Print all subsequences → a contigous / non-contigous sequences, which follows the order.

$n = 3$

arr → $\{3, 1, 2\}$

3 2 1

$\{\}$ → ✓

3 ✓

1 ✓

2 ✓

3 1 ✓

1 2 ✓

3 2 ✓

3 1 2 ✓

→ 8

```
f(ind, [])
{
    if (ind >= n)
        print([])
        return;

    [].add(arr[i]);

    f(ind+1, []); -> take

    [].remove(arr[i]);

    f(ind+1, []); -> not take

}
```

x x x {}

arr → [3, 1, 2]

n=3

```
f(ind, [])
{
    if (ind >= n) ×
        print([])
        return;
    [].add(arr[i]);          [3]
    f(ind+1, []); → take
    [].remove(arr[i]);
    f(ind+1, []); → not take
}

main()
{
    arr → {3,1,2}
    f(0,[])
}
```

```
f(1, [3])
{
    if () ×
    [3,1]  [3].add(arr[i])
    f[2, [3,1]]
    [3]  .remove
    f(2, [3])
```

```
f(2, [3,1])
{
    if ×
    [3,1,2]  [3,1].add(arr[2])
    f(3, [3,1,2])
    [3,1] ← [3,1,2].remove(arr[2])
    f(3, [3,1])
}
```

```
f(3, [3,1,2])
{
    if (ind >= n) ✓
        print ✓
        return.
```

```
f(3, [3,1])
{
    if (ind >= 3) ✓
        print
        ret
```

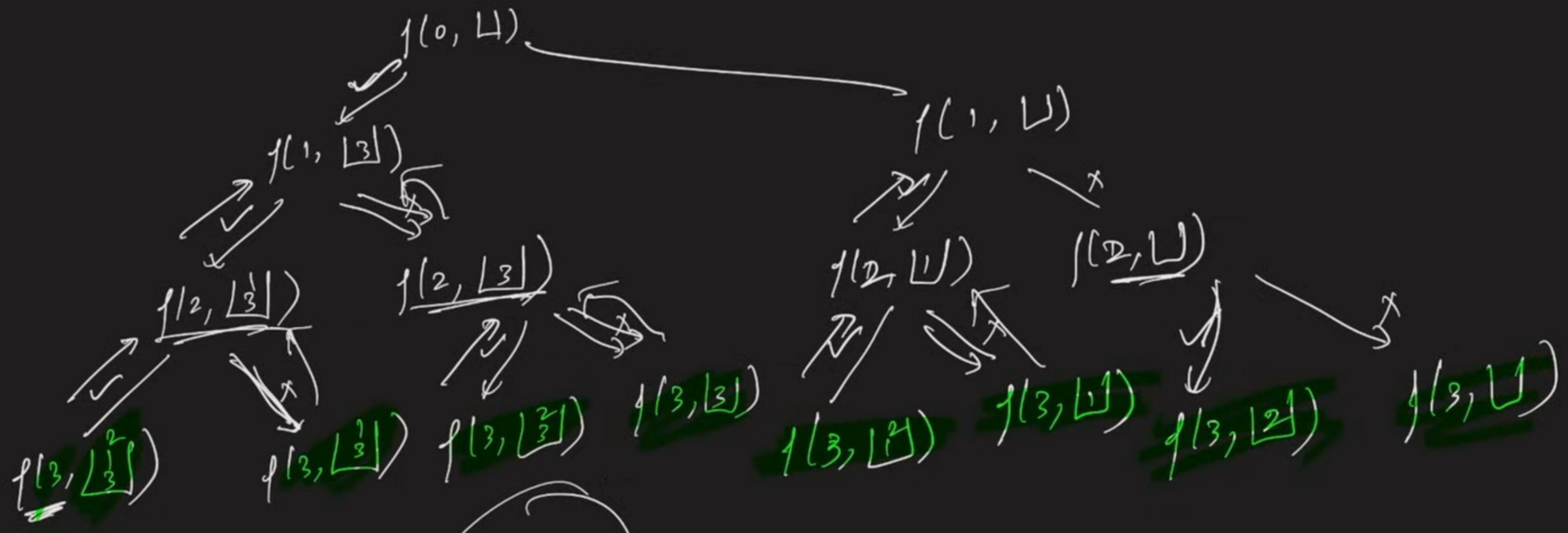
f(2, [3])

3 = take → 3,1 ; 3 → [3]

3 x — ✓ → (3, 2) ; x → (3)

L6. Recursion on Subsequences | Printing Subsequences

2.30



2 1 2
3 1
3 2
3

4

1 2
1
2
{ }

4

19:09 / 25:00

## L6. Recursion on Subsequences | Printing Subsequences

code.cpp   stdc++.h

```cpp
#include<bits/stdc++.h>
using namespace std;
void printF(int ind, vector<int> &ds, int arr[], int n) {
    if(ind == n) {
        for(auto it : ds) {
            cout << it << " ";
        }
        if(ds.size() == 0) {
            cout << "{}";
        }
        cout << endl;
        return;
    }
    // take or pick the particular index into the subsequence
    ds.push_back(arr[ind]);
    printF(ind+1, ds, arr, n);
    ds.pop_back();

    // not pick, or not take condition, this elemnt is not added to your subsequenc
    printF(ind+1, ds, arr, n);
}
int main() {
    #ifndef ONLINE_JUDGE
    freopen("input.txt", "r", stdin);
    freopen("output.txt", "w", stdout);
    #endif
    int arr[] = {3, 1, 2};
    int n = 3;
    vector<int> ds;
    printF(0, ds, arr, n);
```

input.txt

```
5
1 2 3 4 5
```

output.txt

```
3 1 2
3 1
3 2
3
1 2
1
2
{}
```

```
[Finished in 1.1s]
```

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